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Course: Data Science
Exercise 6: Date_ Functions

1. Select emp_id
name,
~~hiredate~~ hire_date,
DATEADD (Month, 6, hire_date) AS hire_plus_6_months
From Employees;

| emp_id | name | hire_date | hire_plus_6_months |
|---|---|---|---|
| 1 | Alice | 2020-01-15 | 2020-07-15 |
| 2 | Bob | 2021-06-10 | 2021-12-10 |
| 3 | Charlie | 2023-03-22 | 2023-09-22 |

2. Select student_id,
name,
DATEDIFF (Day, dob, '2026-04-19') AS age_in_days
From Students;

| student_id | name | age_in_days |
|---|---|---|
| 101 | Maya | 7557 |
| 102 | Ethan | 7809 |
| 103 | Sienna | 7340 |

3. Select event_id,
event_name
DATEDIFF (Day, '2026-04-19', event_date) AS days_remaining
From Events;

3.

| event_id | event_name | days_remaining |
|---|---|---|
| 1 | Seminar | -673 |
| 2 | Workshop | -230 |
| 3 | Hackathon | -454 |

Since it shows negative, then it means the event has already passed.

4. Select invoice_id,
issue_id,
due_id,
DATEDIFF(Day, issue_id, due_id) AS days_between
From Invoices;

| invoice_id | issue_id | due_id | days_between |
|---|---|---|---|
| 501 | 2025-03-10 | 2025-03-25 | 15 |
| 502 | 2025-04-01 | 2025-04-15 | 14 |
| 503 | 2025-04-10 | 2025-04-20 | 10 |

5. Select course_id,
name,
TO_CHAR(start_date, 'mmmm yyyy') AS formatted_date
From Courses;

| course_id | name |
|---|---|
| 501 | SQL Basics |
| 502 | |
| 503 | |

| course_id | name | formatted_date |
|---|---|---|
| 201 | SQL Basics | May 2025 |
| 202 | Python | June 2025 |

6. Select member_id,
DATE_FROM_PARTS(start_year, start_month, start_day) AS full_start_date
From Memberships

| member_id | full_start_date |
|---|---|
| 1 | 2023-05-10 |
| 2 | 2022-11-25 |

7. Select sub_id,
plan,
DATEADD(Year, 1, renewal_date) AS extended_renewal_date
From Subscriptions

| sub_id | plan | extended_renewal_date |
|---|---|---|
| 11 | Basic | 2026-01-01 |
| 12 | Premium | 2026-03-15 |

8. Select order_id,
order_date,
TO_DATE(CURRENT_DATE) AS today_date,
DATEDIFF(Day, order_date, CURRENT_DATE) AS days_since_order
From Orders;

| order_id | order_date | today_date | days_since_order |
|---|---|---|---|
| 1001 | 2025-04-15 | 2026-04-19 | 368 |
| 1002 | 2025-04-10 | 2026-04-19 | 373 |

9. Select training_id,
topic,
DATE_PART (Year, training_date) AS training_year
From Trainings;

| training_id | topic | training_year |
|---|---|---|
| 1 | Safety | 2025 |
| 2 | Compliance | 2025 |

10. Select post_id,
title,
DATE_PART (Hour, published_on) AS hour_published
DATE_PART (Minute, published_on) AS minute_published
From Blog_Posts;

| post_id | title | hour_published | minute_published |
|---|---|---|---|
| 1 | SQL Tips | 10 | 15 |
| 2 | Data Cleaning | 16 | 45 |

11. Select driver_id,
~~lices~~ license_expiry,
DATEDIFF (Day, CURRENT_DATE, license_expiry) AS days_left
From Drivers;

| driver_id | license_expiry | days_left |
|---|---|---|
| 301 | 2025-08-10 | -252 |
| 302 | 2023-12-31 | -840 |

License has already expired, hence the negative results

12. Select message_id,
sent_timestamp,
CURRENT_TIMESTAMP AS current_timestamp,
DATEDIFF(Second, sent_timestamp, CURRENT_TIMESTAMP) AS
seconds_since_sent
From Messages;

| message_id | sent_timestamp | current_timestamp | seconds_since_sent |
|---|---|---|---|
| 1 | 2025-04-19 09:32:45 | 2026-04-19 15:02:51 | 31555806 |
| 2 | 2025-04-18 23:59:59 | 2026-04-19 15:02:51 | 31590172 |

13. Select return_id,
return_date,
DATEADD(Day, 15, return_date) AS restock_date
From Returns;

| return_id | ~~rest~~ return_date | restock_date |
|---|---|---|
| 901 | 2025-04-05 | 2025-04-20 |
| 902 | 2025-04-01 | 2025-04-16 |

14. Select assign_id,
TO_DATE(assigned_on, 'YYYY-MM-DD') AS assigned_on_date
From Assignments;

| assign_id | assigned_on_date |
|---|---|
| 1 | 2025-03-01 |
| 2 | 2025-03-05 |

15. Select meeting_id,
TO_CHAR( scheduled_time, 'MMMM dd, yyyy "at" hh:mm AM')
AS formatted_meeting_time
From Meetings;

| meeting_id | formatted_meeting_time |
|---|---|
| 1 | April 19, 2025 at 02:00 PM |
| 2 | April 19, 2025 at 09:30 AM |